

BAREDDY MADHU SUDHAN REDDY

AI Engineer | Python | Machine Learning | Scikit-learn | Pandas | NumPy | SQL

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PROFESSIONAL SUMMARY

Aspiring AI Engineer and Computer Science graduate with a strong foundation in Python, machine learning, data processing, SQL, and backend development. Hands-on experience building an unsupervised machine learning project using K-Means and Isolation Forest with Python, scikit-learn, Pandas, NumPy, and Matplotlib. Experienced in data preprocessing, data cleaning, exploratory data analysis, model evaluation, visualization, debugging, testing, and software development practices. Also experienced in Python backend development through internship work involving modular programming, REST APIs, testing, debugging, and Agile/Scrum.

TECHNICAL SKILLS

Programming: Python, JavaScript

AI & Machine Learning: Machine Learning, Unsupervised Learning, K-Means Clustering, Isolation Forest, Anomaly Detection, Model Evaluation, Precision, Recall

Data Science: Pandas, NumPy, Matplotlib, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Data Visualization

Backend: Django, Flask, FastAPI, REST APIs, JSON, API Integration

Databases: SQL, MySQL, PostgreSQL, CRUD, Relational Database Design, Query Optimization

Software Engineering: OOP, DSA, SDLC, Clean Code, Debugging, Code Review, Documentation

Testing: pytest, unittest, Unit Testing, Functional Testing, Code Coverage

Tools: Git, GitHub, GitLab, VS Code, PyCharm, Linux/Ubuntu, Docker, Docker Compose

Methodologies: Agile, Scrum, Sprint Planning

PROFESSIONAL EXPERIENCE

Python Developer Intern | SkillRaace

Dec 2025 – Mar 2026

- Developed reusable Python modules using object-oriented programming principles.
- Worked on software development, debugging, testing, code review, and documentation.
- Resolved 10+ production bugs through log analysis and root-cause debugging.
- Developed automation scripts and unit/functional tests using pytest and unittest.
- Improved code coverage and software reliability through systematic testing.
- Participated in Agile/Scrum development cycles and technical discussions.
- Contributed to backend solution design and implementation.

AI/ML PROJECT

Anomaly Detection in Railway Station Safety Using Unsupervised Machine Learning | Python, scikit-learn, Pandas, NumPy, Matplotlib

- Developed an anomaly detection solution for railway station safety using unsupervised machine learning techniques, applying K-Means clustering and Isolation Forest to identify patterns and flag anomalous observations.
- Processed and cleaned raw data using Pandas and NumPy, and performed exploratory data analysis to understand patterns and potential anomalies.
- Created visualizations and analytical dashboards using Matplotlib, and evaluated model results using precision and recall metrics.
- Followed clean coding and software development best practices throughout the project.

ADDITIONAL PROJECTS

Student Management System | Python, OOP, PostgreSQL, MySQL

- Developed a modular Python application with CRUD functionality, designing relational database structures with validation and data-integrity checks.
- Applied OOP principles for maintainable application development.

Retail Sales Data Processing & Analysis | Python, SQL, PostgreSQL, Pandas, NumPy, Matplotlib

- Processed 10,000+ sales records using Python and SQL, performing data cleaning, transformation, analysis, and visualization.
- Improved SQL query execution time by approximately 20% and reduced data inconsistencies by approximately 30%.

EDUCATION

B.Tech, Computer Science & Engineering — Sri Venkatesa Perumal College of Engineering & Technology, Puttur, AP

2022 – 2026 | CGPA: 7.9 / 10

Intermediate (MPC) — Nandyal Junior College, Nandyal, AP

2020 – 2022 | CGPA: 7.35 / 10

CERTIFICATIONS

- Artificial Intelligence — NPTEL
- Python Programming — Skills4India
- HTML & CSS — Great Learning